Practical assignment AE4317

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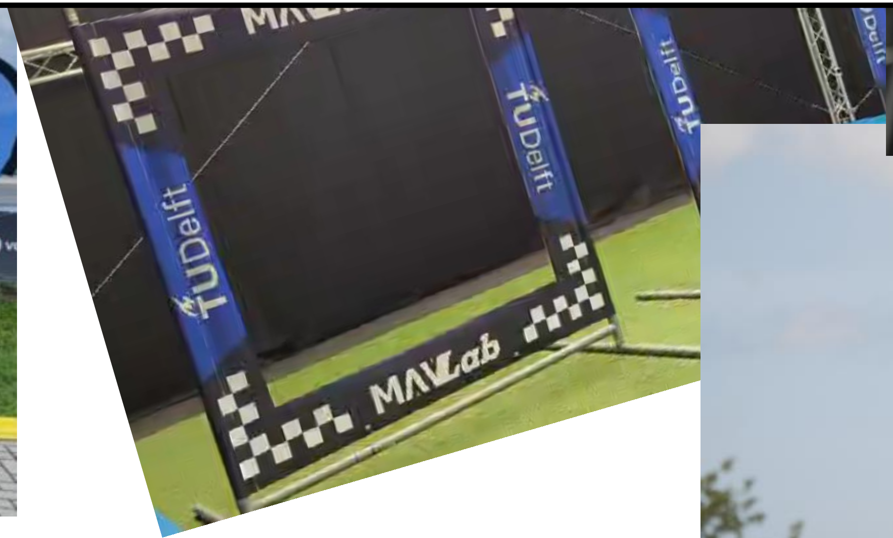
Why?

IDEA into Robot



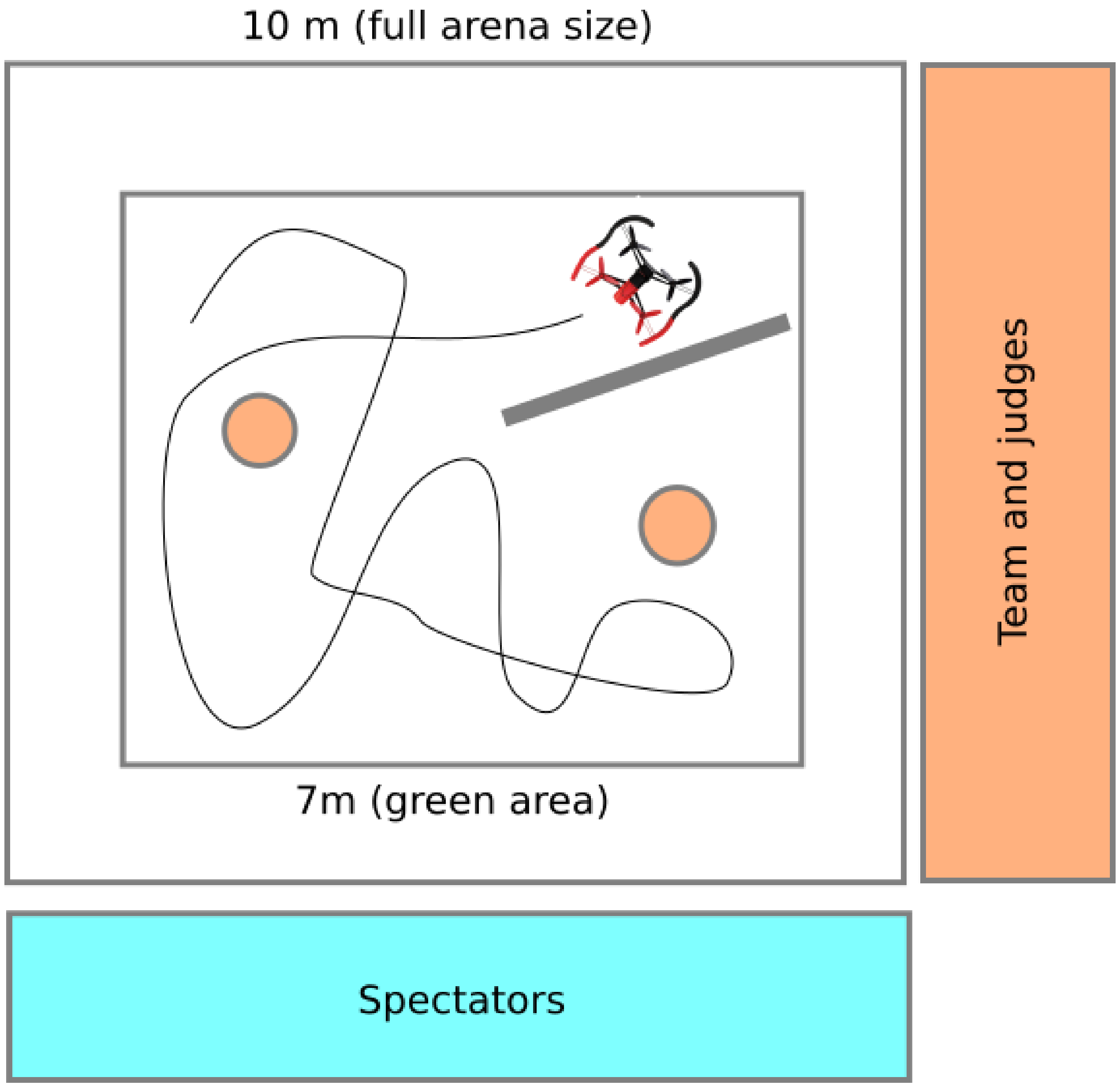
Practical assignment

Objective:

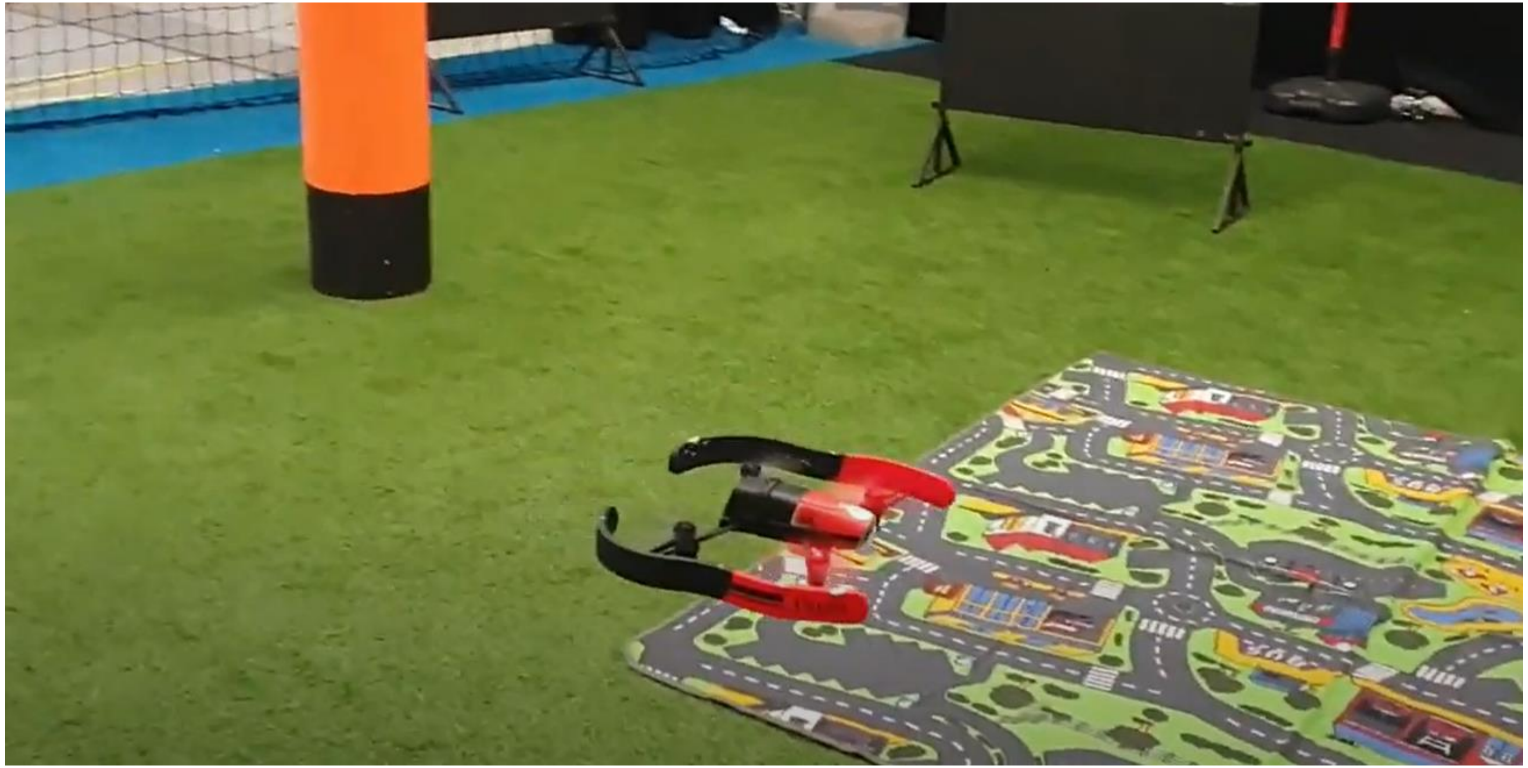
- Students can apply the learned AI and control techniques to, and create new algorithms for an autonomous flight task.
- Students can perform this task as independently as possible – the solution is not provided in the theoretical course; the practical session lectures only provide a starting point.
- Students will learn how to (better) program in Python and C, and will learn to use GIT.

“The Drone”

"The Task"


10 m (full arena size)
7m (green area)
Team and judges
Spectators

Impression – Competition



https://youtu.be/R3LNWh3UQHk (2015)

https://youtu.be/T4012y4wEYQ (2024)

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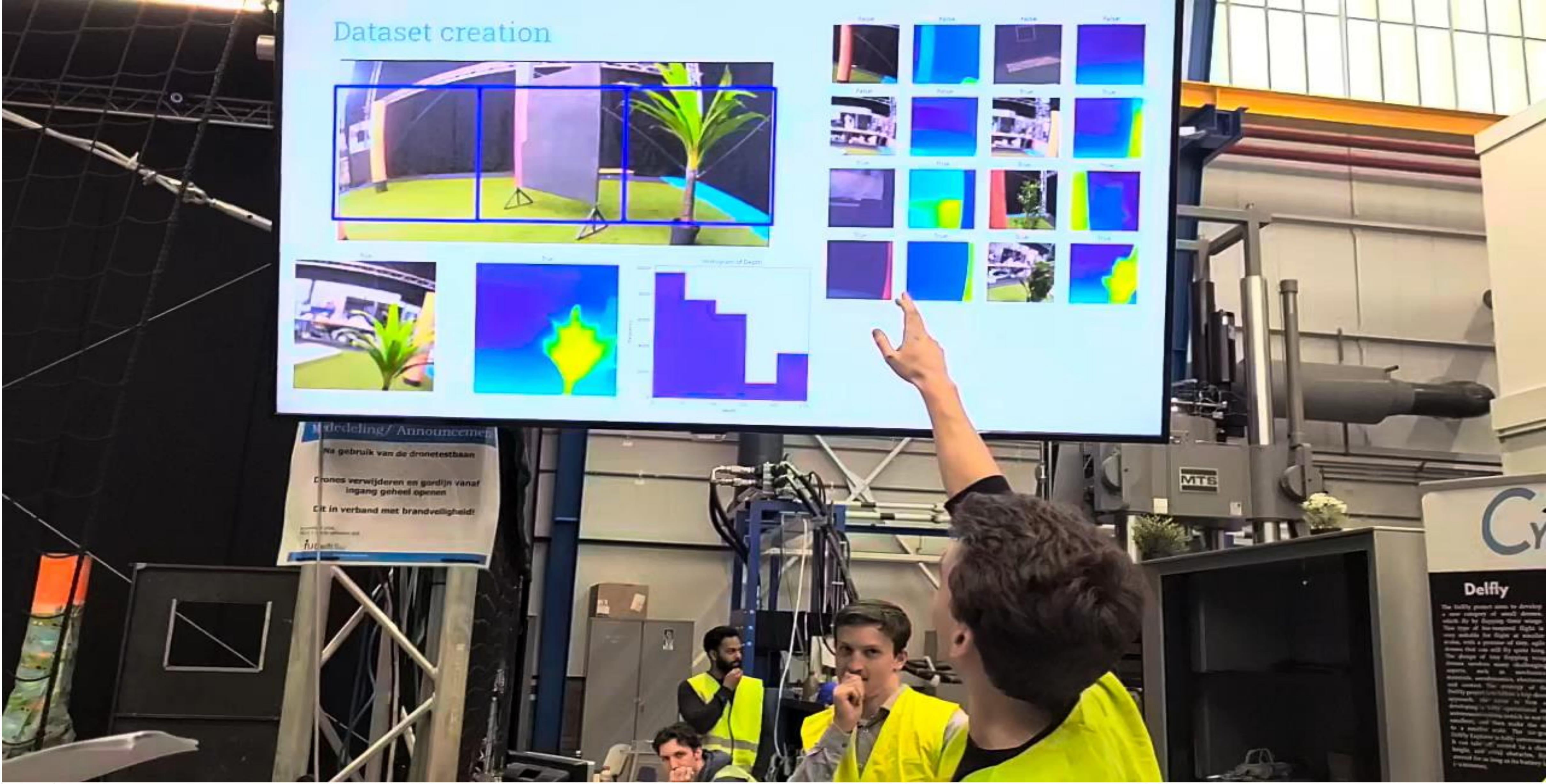
Dataset creation
Mededeling/ Announcemen
Na gebruik van de dronetestbaan
Drones verwijderen en gordijn vanaf ingang geheel openen
Dit in verband met brandveiligheid!
MTS
Delfly

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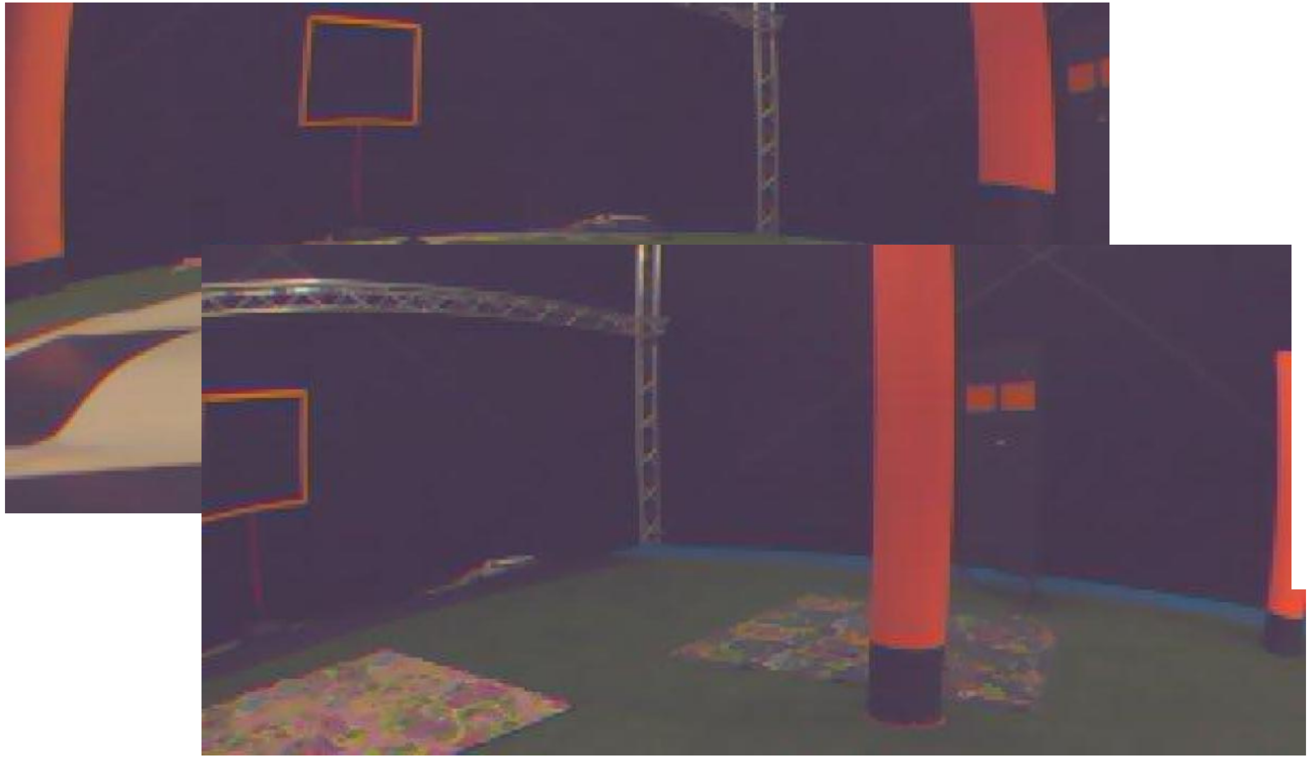
Simulation

Always first implement and test your vision and control algorithms in simulation!

Sample dataset



https://surfdrive.surf.nl/files/index.php/s/68S6moWEIqCT14K

Template planning

	W1	W2	W3	W4	W5	W6	W7
Form groups							
Install paparazzi							
Try out provided orange module							
All crash course modules							
Devise strategy							
Prototype vision algorithms (scripts / sim)							
Implement control algos in Paparazzi (sim/real)							
Implement vision algos in Paparazzi (sim/real)							
Test, iterate							
FINALIZE CODE FOR COMPETITION							D-DAY

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CyberZoo

Access to the hall is regulated.

- Develop in simulation
- First week of test flying:
  - February 28th
- **Competition**
  - **March 28th**

Planning 2025

February 12, CZ-B:	Introduction (G)
February 14, CZ-K:	Practical I (Paparazzi – Linux) (C)
February 19, CZ-B:	Computer vision I – Monocular obstacle detection (C)
February 21, CZ-K:	Practical (Crash course C / GIT) (C)
February 26, CZ-D:	Computer vision II – Computational effort (C)
February 28 1, CZ-K:	Practical
March 5, CZ-D:	Computer vision III – stereo vision (C) / Practical
March 7, CZ-K:	Practical
March 12, CZ-D:	Optical flow I (G)
March 14, CZ-K:	Practical
March 19, CZ-D:	Optical flow II (C)
March 21, CZ-K:	Practical
March 26, CZ-D:	Exam practice (C)
March 28, Cyberzoo:	**Competition, 9:00 – 12:30**
March 31, Brightspace:	**Deadline group report: Assignments**
April 9:	**9-12h – EXAM**